

Iteration 05 – V&V Report

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Project: Firefighting Drone Swarm Simulation

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1. Overview

This document is the Verification and Validation (V&V) report for Iteration 05 of the Firefighting Drone Swarm Simulation. It maps every test requirement defined in the V&V specification (and the finalized Metric Requirements from the course discussion forum) to a concrete, executable JUnit test and explains how pass criteria are verified. The full test suite is invoked with:

```
`mvn test`
```

✓ Total tests: 98 | Failures: 0 | Skipped: 0

2. New Capabilities Verified in Iteration 05

Capability	Where implemented
Battery drain during flight (% per second)	DroneSubsystem+DroneConfig
Fuel drain per metre travelled	DroneSubsystem+DroneConfig
REFILLINGstate resets water / battery / fuel	DroneSubsystem.performRefill()
Liquid-agent capacity enforcement	DroneSubsystem— agent capped atAGENT_CAPACITY_LITRES
LinkedBlockingQueue task queue (no task loss)	DroneSubsystem.taskQueue
Battery + fuel fields inDroneUpdate	DroneUpdate7-param constructor;PacketBuilder/PacketParser
All 5 required + 2 optional performance metrics	SimulationMetrics(see §5)
File-based logging	SimulationLogger→logs/directory
DroneSubsystemstartup race-condition fix	Receiver startedbeforeREGISTER is sent
SimulationMetricssidempotent fire recording	recordFireDetected/recordFireExtinguisheduseputIfAbsent
FireIncidentSubsystemmalformed-line recovery	Per-linetry/catchinparseIncidentReport
DroneInfothread-visible fields	completedAssignments,zonesServiced, etc. declaredvolatile
RuntimeGUIEDT-safe fault timers	Timer/HashMapmutations viaSwingUtilities.invokeLater

2.1. Final Submission Checklist

- Final Iteration 05 source snapshot is included with runnable Maven project files.
- `mvn test` passes cleanly with the documented test count and no failures.
- V&V traceability tables map unit, integration, and metrics tests to implemented behaviour.
- README's Iteration 05 design summary reflects final shipped behaviour.
- Final demo inputs (event, zone, and fault CSVs) are clearly identified.
- Persistent runtime artifacts are present under `logs/`.
- Final report, README, diagrams, and test evidence are internally consistent.
- Demo covers normal operation, stuck-flight, nozzle jam, and reassignment.
- All required course artifacts (report, code, tests) are included.

3. Unit Tests

3.1. Drone Subsystem

Test ID	Class / Method	Scope	Pass Criteria
UT-DRONE-STATE-01	testIdleToArrivedTransition	IDLE → EN_ROUTE → ARRIVED	State =ARRIVED
UT-DRONE-STATE-02	testFullStateMachineCycle	Full cycle through all states	All expected states observed
UT-DRONE-CAPACITY-01	testWaterDepletedThenRefilled	Agent depletion and restoration	Water = 0 then water = 15 L
UT-DRONE-CAPACITY-02	testRefillBetweenMissions	Multiple queued tasks	Refill observed between missions
UT-DRONE-FAULTFLAG-01	nozzleStuckClosedFaultFlagTest	Soft fault: nozzle stuck closed	Drone recovers toIDLE
UT-DRONE-FAULTFLAG-02	nozzleStuckOpenHardFaultTest	Hard fault: nozzle stuck open	No subsequentIDLEupdate
UT-DRONE-METRICS-01	droneStartsWithFullConsumables	Initial levels at 100%	Assertions confirm full levels
UT-DRONE-METRICS-02	batteryDepletedDuringFlight	Battery/fuel drain	Levels decrease after flight
UT-DRONE-METRICS-03	waterDepletedAfterDrop	Agent drain after drop	Water = 0 afterDROPPING_AGENT
UT-DRONE-METRICS-04	refillingRestoresAll	Full reset after refill	All values restored to 100%

3.2. Scheduler Subsystem

Test ID	Class / Method	Scope	Pass Criteria
UT-SCHED-QUEUE-01	incidentsServedInArrivalOrder	FIFO arrival order	Zones 1 → 2 → 3 assigned
UT-SCHED-QUEUE-02	schedulerSelectsLessLoaded	Load distribution between 2 drones	Drone 2 selected after Drone 1
UT-SCHED-QUEUE-03	repeatedZoneIncidentsWait	Queueing same-zone fires	Second fire waits for first

Test ID	Class / Method	Scope	Pass Criteria
UT-SCHED-ASSIGN-01	testLoadBalancingPrefersLeastLoaded	Selection based on zones serviced	Less-loaded drone chosen
UT-SCHED-FAULT-01	testDroneStuckTimeout	Timeout logic for stuck drones	Backup drone receives task
UT-SCHED-FAULT-02	testFaultReportTriggersReassignment	Soft fault reassignment	Task reassigned withNONEfault
UT-SCHED-FAULT-03	testHardFaultMarksDroneOffline	Drone removal after hard fault	Healthy drone takes over
UT-SCHED-FAULT-04	testFaultTypeForwarded	CSV toAssignTaskpropagation	faultTypematches CSV
...-05, 06, 07	test...PacketRoundTrip	Data integrity in UDP packets	Fields identical after round-trip
UT-SCHED-FAULT-08	testIncidentReportDefaults	Missing fault field handling	Defaults toNONE
UT-SCHED-PATH-01	testPassesThroughZone_onPath	Redirect logic (on path)	Returnstrue
UT-SCHED-PATH-02	testPassesThroughZone_offPath	Redirect logic (off path)	Returnsfalse
UT-SCHED-PATH-03	testPassesThroughZone_behind	Redirect logic (behind drone)	Returnsfalse
UT-SCHED-PATH-04	testPassesThroughZone_atTarget	Redirect logic (at destination)	Returnsfalse

3.3. Fire Incident Subsystem

Test ID	Class / Method	Scope	Pass Criteria
UT-FIRE-CSV-01	testSendsIncidentReportsFromCsv	CSV parsing logic	UDP fields match CSV input
UT-FIRE-CSV-02	testEmptyFileProducesNoPackets	Empty input handling	No packets sent in 2s
UT-FIRE-CSV-03	testSingleEventFile	Single row CSV	Correct ID/Severity received

Test ID	Class / Method	Scope	Pass Criteria
UT-FIRE-CSV-04	testMalformedLineSkipped	Resilience to bad CSV rows	Processing continues after skip
UT-FIRE-ZONE-01	ZoneCsvParserTest.*	Zone file parsing suites	Valid zones load correctly

3.4. UDP and Communication Utilities

Test ID	Class / Method	Scope	Pass Criteria
UT-NET-PKT-01	PacketRoundTripTest.*	All packet types round-trip	Preserved exactly
UT-NET-PKT-02	droneUpdateRoundTripIncludesMetrics	Battery/Fuel in packets	Fields preserved
UT-NET-PKT-03	legacyDroneUpdateDefaults	Backward compatibility	Defaults to 100%
...-LOSS-01...11	PacketParserTest.test...	Resilience to malformed/empty bytes	Returns null, no crashes

3.5. Scheduler Logic and Fairness

Test ID	Class / Method	Scope	Pass Criteria
UT-SCHED-SEL-01	testAssignsDroneToIncident	Basic assignment	AssignTask received
UT-SCHED-SEL-02	testHighSeverityAssignment	Severity priority handling	Correct severity carried
UT-SCHED-SEL-03	testTwoDronesFairness	Fairness between two drones	No duplicate assignments
UT-SCHED-BAL-01	balancesAssignmentsAcrossMultiple	Proximity + Load balancing	Drones assigned to nearest zones

3.6. Simulation Metrics

Test ID	Class / Method	Scope	Pass Criteria
UT-METRICS-DEDUP-01	duplicateFireDetectedDoesNotOverCount	Idempotent detection	Counter stays at 1
UT-METRICS-DEDUP-02	duplicateFireExtinguishedDoesNotOverCount	Idempotent extinguish	Counter stays at 1

Test ID	Class / Method	Scope	Pass Criteria
UT-METRICS-COUNT-01	distinctZonesCountedSeparately	Unique zone counting	Distinct totals correct

3.7. Domain Model

Test ID	Class / Method	Pass Criteria
UT-MODEL-SEV-01	testLowRequiresTenLitres	getReqLitresOfWater()= 10
UT-MODEL-SEV-02	testModerateRequiresTwenty	getReqLitresOfWater()= 20
UT-MODEL-SEV-03	testHighRequiresThirty	getReqLitresOfWater()= 30
UT-MODEL-SEV-04	testAllSeverityValuesPresent	Enum covers LOW/MOD/HIGH
UT-MODEL-SEV-05	testRequiredLitresOrdered	LOW < MODERATE < HIGH
UT-MODEL-TIMER-01	testOneShotTimer	Fires callback exactly once
UT-MODEL-TIMER-02	testPeriodicTimer	Fires at regular intervals
UT-MODEL-TIMER-03	testDefaultConstructor	Runs without error

3.8. Messaging DTOs

Test ID	Class / Method	Scope	Pass Criteria
UT-DTO-IR-01	testAllFieldsPreservedLow	LOW severity round-trip	Fields preserved
UT-DTO-IR-02	testAllFieldsPreservedHigh	HIGH severity round-trip	Fields preserved
UT-DTO-IR-03	testAllFieldsPreservedModerate	MODERATE round-trip	Fields preserved
UT-DTO-IR-04	testPacketTypeIsIncidentReport	Header check	Correct type returned

3.9. GUI State Tracking

Test ID	Class / Method	Scope	Pass Criteria
UT-GUI-01	testDroneStateTracking	Mission cycle states	Reflects updates
UT-GUI-02	testDronePositionTracking	X, Y position updates	Reflects updates

Test ID	Class / Method	Scope	Pass Criteria
UT-GUI-03	testDroneWaterTracking	Agent levels in GUI	Reflects updates
UT-GUI-04	testActiveFireCountTracking	Fire tally accuracy	Correct increment/decrement
UT-GUI-05	testEventLogging	Accumulating log entries	Entries found in all Logs
UT-GUI-06	testClearFireTile	UI fire removal	Counter increments

4. Integration Tests

Test ID	Class	Description	Pass Criteria
IT-01	MissionLifecycleTest	Single drone Nominal	Full IDLE→COMPLETED cycle
IT-02	MultiIncidentQueueDrainIT	3 drones, 6 incidents	No task loss
IT-03	MultiIncidentLoadBalanceIT	Mixed severities	Balanced load
IT-04	FaultHandlingIT	Fault injection via CSV	Reassignment detected
IT-05	UdpLoopbackIT	Network contracts	Packet contents preserved
IT-06	TwoDronesBasicFlowIT	End-to-end multi-drone	Successful mission
IT-07	testDroneQueuesMultiple	In-drone task queueing	Serial mission completion
IT-08	testLoadBalancingLeastLoaded	Selection logic verification	Correct drone chosen
IT-09	testPathThroughRedirect	Dynamic redirect on path	Original zone re-queued

5. Performance Metrics Tests

SimulationMetricscomputes all 7 required metrics.

#	Metric Name	Method	Validation
1	Avg Response Time	getAverageResponseTimeMs()	Created → ARRIVED
2	Max Response Time	getMaxResponseTimeMs()	Per-fire maximum
3	Avg Completion Time	getAverageExtinguishTimeMs()	Created → Extinguished
4	Max Completion Time	getMaxExtinguishTimeMs()	Per-fire maximum
5	Utilization %	getUtilizationPct()	Reported in logs
6	Avg Queue Length	getAverageQueueLength()	Sampled during ops
7	Max Queue Length	getMaxQueueLength()	High-water mark

6. Logging

SimulationLoggerwrites five persistent files to thelogs/directory:

- `events.log`: High-level scheduler events.
- `telemetry.log`: Periodic drone status (battery, fuel, pos).
- `assignments.log`: Task assignment + completion history.
- `queue.log`: Incident-queue depth changes.
- `metrics.log`: Final performance summary string.

7. System and Acceptance Test Status

Test ID	Description	Status
ST-01	Nominal full-scale (10 drones)	Manual Passed
ST-02	Fault-heavy scenario	Auto Covered
ST-03	Overload & overflow resilience	Auto Covered
ST-04	Timing & instrumentation	Auto Covered
ST-05	GUI & Usability mapping	Manual Passed

8. Port Registry

Integration tests use assigned ports outside the production range (6000) for isolation.

Range	Assigned To
15201 – 15312	Drone State & Capacity Tests
15401 – 15404	Fire Incident Subsystem
15501 – 15554	Scheduler Selection Tests
15700 – 15756	Load Balancing Tests
15800 – 15955	Lifecycle & Queue ITs
16100 – 16452	Fault Handling Tests
16500 – 16560	Fault Recovery & Metrics
16600 – 16752	Queue Order & Selection

9. Known Limitations

- ST-01 and ST-05 require displays (validated manually).
- Drone metrics rely on OS schedulers for UDP delivery; race-condition fix ensures stability.
- Inter-event sleep (2s) in the subsystem dominates playback metrics.
- Occasional flakiness in load balance tests under high CPU load; mitigated by socket retries.